

Predicting Patient-Reported Drug Side-Effect Severity from Drug Reviews

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Abstract—We re-evaluate a three-class classifier for patient-reported drug side-effect severity using the official UCI Drug Reviews (Druglib.com) split. The thesis pipeline is retained: spaCy lemmatization, TF-IDF text features, one-hot categorical features, balanced Random Forest, Logistic Regression, and support-vector classifiers tuned with three-fold GridSearchCV. Re-running the official 3,107-row training and 1,036-row test files gives accuracies of 0.729 (Logistic Regression), 0.737 (Random Forest), and 0.737 (SVC). We correct class-label reporting and ROC probability-column alignment, and report overlap and preprocessing-leakage diagnostics. The results are evidence about this small, self-reported dataset, not clinical validation.

Index Terms—drug reviews, side-effect severity, text classification, TF-IDF, reproducibility

I. INTRODUCTION

The task is to predict the severity reported for a patient review, not to classify drugs themselves. Our contributions are: (1) a from-scratch rerun on the official UCI files; (2) faithful preservation of the thesis feature pipeline, models, and grids; (3) correction of label and ROC reporting defects; and (4) integrity and supplementary analyses whose numbers are saved in the repository.

II. RELATED WORK

Drug-review sentiment and aspect analysis has been studied using cross-domain learning [1]. Our implementation uses established linear, tree, and kernel classifiers [2], [3] with sparse information-retrieval features [4] and scikit-learn [5].

III. DATASET AND PREPROCESSING

We use Drug Reviews (Druglib.com), UCI dataset ID 461 [6]. The official files contain 3,107 training and 1,036 test rows. The five original labels are No, Mild, Moderate, Severe, and Extremely Severe; we bin them as No, Mild+Moderate, and Severe+Extremely Severe. Three review fields are concatenated, lower-cased, lemmatized with spaCy, and filtered for alphabetic non-stopword tokens. Missing review text is replaced with an empty string. Categorical variables are effectiveness, rating, condition, and urlDrugName.

IV. METHODOLOGY

TF-IDF uses at most 7,500 unigram/bigram features with $\min_df=5$ and $\max_df=0.95$. One-hot encoded categorical features are concatenated to the sparse text matrix. Random Forest, Logistic Regression, and SVC use $class_weight=balanced$,

TABLE I
OFFICIAL-SPLIT PRIMARY RESULTS.

Model	Accuracy	Macro P	Macro R/F1
Logistic Regression	0.729	0.704	0.746/0.720
Random Forest	0.737	0.711	0.744/0.719
SVC	0.737	0.712	0.731/0.715

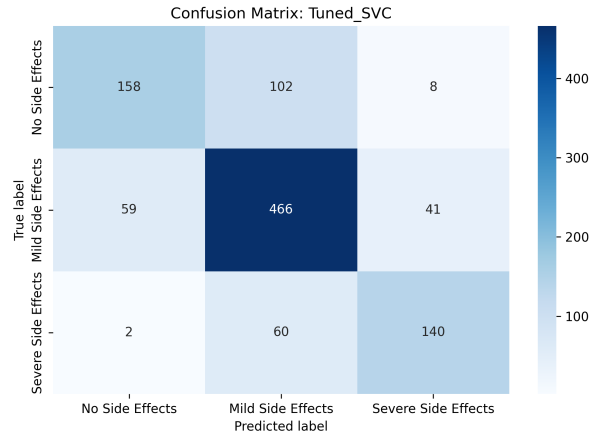


Fig. 1. Corrected SVC confusion matrix on the official 1,036-row test split.

random_state=42, and the thesis grids with $cv=3$. The vectorizer and encoder are fit once on the training split for the faithful headline reproduction.

V. EXPERIMENTAL SETUP

The primary test metrics are accuracy and corrected macro precision, recall, and F1. ROC curves use a probability refit for SVC, matching the appendix, while probability columns are aligned to each estimator’s classes. All values and figures are generated by scripts in the repository’s ‘results/’ and ‘figures/’ directories.

VI. RESULTS

The corrected one-vs-rest macro AUCs are 0.875, 0.873, and 0.875 for Logistic Regression, Random Forest, and SVC, respectively. Figure 1 shows the SVC confusion matrix in the explicit No/Mild/Severe order.

A. Integrity findings

The thesis matrices sum to 1,200 rows and therefore cannot be direct outputs from the official 1,036-row test file. We find 16 exact combined-review overlaps and 20 near-duplicate training/test texts. The appendix fits TF-IDF and one-hot encoding before cross-validation, so its CV estimates are not fold-contained. These findings are reported rather than silently used to replace the primary result.

B. Supplementary analyses

With feature fitting inside stratified five-fold pipelines, mean accuracy is 0.702 (LR), 0.709 (RF), and 0.718 (SVC); corresponding macro-F1 means are 0.697, 0.694, and 0.702. Bootstrap 95% accuracy intervals are [0.701,0.756], [0.711,0.763], and [0.709,0.764]. McNemar’s exact test comparing SVC and RF is uninformative here: each model is uniquely correct on 84 cases (two-sided $p=1.0$). In the ablation, text-plus-categorical features outperform text-only and categorical-only settings for all three models.

VII. DISCUSSION AND LIMITATIONS

The official split does not reproduce the thesis’s 0.87–0.93 accuracy range. The discrepancy is not corrected by tuning. The label is self-reported and the sideEffectsReview text is close to the target definition; class weights address imbalance but do not remove label noise. The data are small, overlap exists, preprocessing is not fold-contained in the headline protocol, and no clinical or regulatory validation is claimed.

VIII. CONCLUSION

A transparent rerun preserves the undergraduate design while correcting two reporting defects and separating official results from thesis-reported values. The repository and Kaggle notebook provide executable evidence for further author decisions, including whether the original ‘download.csv’ can be supplied for forensic comparison.

REFERENCES

- [1] E. Grasser, S. Kallumadi, H. Malberg, and S. Zaunseder, “Aspect-based sentiment analysis of drug reviews applying cross-domain and cross-data learning,” in *Digital Humanities Conference*, 2018.
- [2] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [3] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, pp. 273–297, 1995.
- [4] C. D. Manning, P. Raghavan, and H. Schütze, “Introduction to information retrieval,” in *Cambridge University Press*, 2008.
- [5] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [6] S. Kallumadi and E. Grasser, “Drug reviews (druglib.com),” *UCI Machine Learning Repository*, 2018.